PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: GreenTrust Pulse

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1. Project Overview

- **Problem Statement:** Malaysian SMEs face increasing pressure to comply with Environmental, Social, and Governance (ESG) standards while maintaining financial performance and accessing green financing opportunities. However, their decision-making is hindered by fragmented data sources, limited analytical capabilities, and a lack of integrated, real-time ESG intelligence.

  Despite the availability of low-interest green financing schemes, many SMEs fail to qualify due to poor visibility into their ESG performance, risk exposure, and eligibility criteria. Current ESG assessment processes are manual and delayed, preventing SMEs from detecting supply chain risks, responding to external events, and understanding how sustainability efforts impact financial outcomes.

  As a result, SMEs rely on reactive and intuition-based decisions, leading to missed financing opportunities, higher operational risks, and reduced competitiveness in ESG-driven supply chains. Furthermore, as multinational corporations (MNCs) increasingly prioritise ESG-compliant suppliers, SMEs without verifiable ESG performance risk losing key contracts and weakening their overall financial position.

  There is a clear need for a system that provides continuous, explainable, and data-driven ESG insights to support timely decision-making, improve financing readiness, and align sustainability performance with financial outcomes.

- **Target Domain:** Intelligent ESG-driven financial decision systems that enable SMEs to continuously monitor their ESG performance, derive actionable insights from AI analysis, and execute data-driven decisions in financing, operations, and cost optimization.

- **Proposed Solution Summary:** GreenTrust Pulse is an AI-powered, cloud-based ESG decision intelligence platform designed to help SMEs overcome fragmented data and limited visibility by providing real-time, integrated, and explainable ESG insights. Powered by Z.ai GLM as a core LLM service layer and supported by continuous data ingestion (24/7 streaming processing), the system ensures that ESG information is always up-to-date and contextually analysed. It combines GraphRAG-based knowledge graph reasoning with a multi-agent framework (ESG auditor, CFO, fraud detector, and arbitrageur) to analyse ESG performance from multiple perspectives and ensure reliable, cross-validated outputs.

  By continuously monitoring ESG data across financial records, supplier networks, and external sources, the platform enables early detection of risks, including supply chain disruptions and potential greenwashing. It further supports decision-making through explainable ESG score breakdowns, ESG loan eligibility evaluation, financing recommendations, and predictive forecasting of ESG and financial outcomes.

  Additionally, GreenTrust Pulse includes a what-if simulation feature that allows SMEs to evaluate the impact of operational or financial changes before implementation, as well as automated generation of banker-ready ESG reports for financing applications. Overall, the system provides a comprehensive, data-driven solution that improves ESG transparency, enhances financing readiness, and supports sustainable business growth.

2. Background & Business Objective

- **Background of the Problem:** For many SMEs in Malaysia, decision-making around sustainability, financing, and operations remains fragmented and largely manual. This challenge is significant given that Malaysia has over 1.2 million SMEs, and government initiatives such as the Green Technology Financing Scheme (GTFS) and ESG-linked loan programmes offer interest rates as low as 2%. However, less than 8% of eligible SMEs successfully qualify for such financing. Business data is dispersed across financial records, supplier information, ESG reports, and external sources such as news and regulatory updates, making it difficult to obtain a complete and timely view of ESG performance.

This gap is not primarily due to poor sustainability practices, but rather a lack of real-time visibility and decision support. SMEs are unable to continuously monitor ESG performance, identify risk exposure from suppliers, detect the impact of external events, or translate sustainability efforts into actionable financial decisions.

As a result, business decisions are often based on intuition or delayed insights, limiting the ability to respond to ESG risks and optimise financing opportunities. This issue is further compounded by the time lag in loan evaluation processes, where critical ESG changes may occur before decisions are made.

Overall, the absence of integrated, real-time ESG intelligence systems creates a major barrier for SMEs in achieving both sustainability goals and access to green financing.

- **Importance of Solving This Issue:** Addressing this issue is critical to unlocking the full potential of SMEs in Malaysia's transition towards sustainable and ESG-compliant business practices. SMEs form the backbone of the Malaysian economy, yet many are unable to access green financing opportunities due to the lack of real-time ESG visibility, structured analysis, and decision support. Bridging this gap would not only improve financing accessibility but also enable SMEs to reduce operational risks, optimise costs, and make more informed strategic decisions.

  Furthermore, as ESG requirements become increasingly embedded in global supply chains, the ability for SMEs to demonstrate verifiable ESG performance is essential to remain competitive and retain business relationships with larger corporations. Without proper ESG intelligence, SMEs risk losing contracts, facing higher financing costs, and falling behind in sustainability-driven markets.

Solving this issue also benefits financial institutions by providing more transparent, data-driven, and auditable ESG insights, reducing reliance on manual assessments and improving the efficiency and accuracy of loan evaluation processes. Ultimately, an integrated ESG decision intelligence system supports not only individual business growth, but also broader economic sustainability, financial inclusion, and alignment with national and global ESG goals.

- **Strategic Fit / Impact:** GreenTrust Pulse aligns strongly with national and industry priorities to accelerate ESG adoption, digital transformation, and financial inclusion among SMEs. By addressing the gap between ESG performance and access to green financing, the system supports initiatives such as Malaysia's Green Technology Financing Scheme (GTFS) and broader ESG-linked lending frameworks, enabling more SMEs to qualify for sustainable financing.

  From a business perspective, the platform delivers measurable impact by improving financing approval rates, reducing ESG-related risks, and enhancing operational decision-making through real-time, AI-driven insights. SMEs can proactively manage their ESG performance, optimise supplier choices, and translate sustainability efforts into tangible financial outcomes.

  For financial institutions, the system enhances efficiency and trust by providing explainable, auditable ESG intelligence, reducing reliance on manual due diligence and accelerating loan evaluation processes. At an ecosystem level, the platform contributes to stronger ESG compliance across supply chains, improved transparency, and the creation of a more sustainable and competitive SME sector.

Overall, GreenTrust Pulse serves as a strategic enabler that bridges sustainability and finance, driving long-term economic growth while supporting ESG-driven transformation at scale.

3. Product Purpose

3.1. Main Goal of the System: To provide SMEs with an AI-powered ESG decision intelligence platform that transforms fragmented data into real-time, actionable insights, enabling continuous monitoring, risk detection, and intelligent decision-making across sustainability, financing, and operations, while generating banker-ready ESG reports that support direct application for green financing.

3.2. Intended Users (Target Audience):

The primary target users of GreenTrust Pulse are small and medium enterprises (SMEs) in Malaysia, particularly business owners, financial managers, and operations managers who need to make informed decisions related to sustainability, financing, and cost optimization. The system is especially valuable for SMEs that rely on complex supply chains and require real-time visibility into supplier risks, are seeking to improve or maintain their ESG performance, aim to access green financing but lack structured ESG analysis and reporting tools, and face challenges in integrating and interpreting fragmented business and external data.

In addition, the platform also serves financial institutions and loan officers, who require reliable, explainable ESG assessments to support lending decisions. By providing structured ESG scores, greenwashing risk detection, and audit-ready reports, the system reduces manual due diligence and improves confidence in green loan approvals.

The system also supports ESG program managers and fintech partners (e.g., Ant International) by providing real-time portfolio-level ESG monitoring and enabling the transformation of SME sustainability performance into trusted ESG credit intelligence for financial institutions. This allows stakeholders to track SME improvements over time and generate measurable ecosystem impact metrics such as job creation, carbon reduction, and green financing uptake across the portfolio.

4. System Functionalities

4.1. Description: The system is an AI-powered ESG decision intelligence engine powered by Z.ai GLM, integrating continuous data ingestion, GraphRAG knowledge graphs, and multi-agent reasoning, where specialized agents collaboratively generate transparent and explainable decisions. It processes both structured and unstructured data, including financial records, ESG reports, supplier data, and external news, to generate real-time insights for SMEs.

It detects ESG risks, evaluates supply chain impacts, identifies potential greenwashing activities, and dynamically computes ESG scores. The system also supports predictive forecasting, scenario-based what-if simulations, and generates banker-ready ESG reports to support green financing applications. Overall, it enables transparent, explainable, and data-driven decision-making across sustainability, financing, and operational strategy.

4.2. Key Functionalities:

- **Vibe Designing:** Translate natural-language business and ESG requirements into structured AI-generated insights, including ESG risk analysis, ESG score evaluation, predictive forecasting of ESG impact, financing recommendations, and decision outputs.
- **Conversational Editing:** Allows users to interact with the system through chat-based queries to refine ESG analysis, adjust assumptions, simulate scenarios, and request updated financial or sustainability insights in real time.
- **Versioning & Infinite Canvas:** Enables continuous tracking of ESG evaluations and decision outputs, allowing multiple scenario comparisons, iterative refinements, and historical traceability of AI-generated decisions. It also provides visibility into multi-agent reasoning, showing how individual agent perspectives (ESG auditor, CFO, Arbitrageur, fraud detector) contribute.

- **Export- Component Based:** Converts AI-generated insights into structured, banker-ready ESG reports, including loan recommendations, risk analysis, and executive summaries for financing applications.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The system uses Z.ai GLM as the core AI engine for decision intelligence. It was selected for its strong capabilities in multi-step reasoning, structured output generation, and cross-source information synthesis, which are essential for ESG decision-making tasks. These capabilities enable the model to interpret both structured data (such as financial records and ESG scores) and unstructured data (including reports, news, and regulatory text), making it suitable for complex ESG analysis, risk assessment, and decision support within SMEs.

4.3.2. **Prompting Strategy:** The system adopts a multi-step agentic prompting strategy combined with structured and role-based prompting to support complex ESG decision-making tasks. The overall reasoning process is decomposed into sequential stages, including data interpretation, risk analysis, forecasting, scenario-based (what-if) simulation, and decision synthesis. These stages define what the system needs to solve across each ESG decision workflow.

To execute these reasoning stages, a multi-agent prompting framework is employed, where each specialised agent (ESG auditor, CFO, fraud detector, and arbitrageur) is assigned a distinct role, objective, and evaluation criteria. Each agent contributes a different perspective to the same problem, enabling parallel reasoning, cross-validation, and more reliable ESG decision outcomes.

In addition, structured JSON-guided prompting is used to standardise outputs from each agent, ensuring consistency for downstream processing in GraphRAG and UI components. This allows ESG scores, risk evaluations, forecasts, and recommendations to be reliably extracted, combined, and visualised.

This prompting strategy is well-suited for ESG decision-making because it requires both multi-perspective reasoning and explainability across financial, environmental, and operational dimensions. The combination of multi-step reasoning and multi-agent collaboration ensures higher accuracy, transparency, and robustness compared to single-step or zero-shot approaches. All prompting mechanisms are executed within the Z.ai GLM framework, which serves as the underlying reasoning engine for all agentic and structured outputs.

4.3.3. **Context & Input Handling**: The system is designed to handle both structured and unstructured ESG-related inputs, including financial records, ESG reports, regulatory documents, supplier data, and external news articles. To support large and heterogeneous data sources, the system applies a combination of text preprocessing, chunking, and context retrieval mechanisms before passing inputs into the Z.ai GLM reasoning engine.

For oversized inputs, the system enforces a maximum context window limit of approximately 20,000 tokens per request for end-to-end AI processing. Inputs exceeding this limit are not directly rejected; instead, they are processed using a hierarchical chunking strategy, where documents are split into smaller semantically meaningful segments.

These chunks are then embedded and indexed within the GraphRAG knowledge graph, allowing the system to retrieve only the most relevant sections during reasoning. This ensures that Z.ai GLM operates on contextually relevant information rather than raw full-length documents, improving both efficiency and accuracy.

If an input is excessively large or noisy (e.g., redundant logs or unstructured bulk data), the system applies a filtering layer to remove irrelevant content before chunking. In extreme cases where inputs exceed processing feasibility (e.g., malformed or extremely large uploads), the system returns a structured validation error and prompts the user to reduce or refine the input.

This hybrid approach ensures that the system remains scalable while maintaining high-quality reasoning performance for ESG analysis, risk detection, forecasting, and decision-making tasks.

4.3.4. **Fallback & Failure Behavior:** The system implements multiple fallback mechanisms to ensure robustness when Z.ai GLM produces off-topic, incomplete, or hallucinated outputs. A validation layer first checks all model responses against predefined structured JSON schemas to ensure required fields such as ESG scores, risk classifications, forecasts, and recommendations are present and correctly formatted.

If an output fails validation or is detected as low-confidence, the system triggers a retry mechanism, where the same input is reprocessed using refined prompts with stronger constraints, clearer role definitions, and additional context from GraphRAG retrieval.

To further mitigate hallucination, the system performs cross-verification with GraphRAG, where generated claims related to ESG events, supplier risks, or score changes are validated against retrieved knowledge graph data. Outputs that do not align with verified data are flagged or corrected before being presented to the user.

If repeated attempts still fail to produce a reliable response, the system applies a graceful degradation strategy, returning a simplified output indicating insufficient confidence while still providing any verified partial insights available. For interactive features such as scenario-based (what-if) simulation, if the system detects ambiguous, incomplete, or unresolvable user inputs that lead to unreliable outputs, it prompts the user to refine or re-enter the input parameters before reprocessing the request.

This ensures that ESG analysis, risk detection, forecasting, and financial recommendation reports remain reliable, consistent, and explainable for SME decision-making.

5. User Stories & Use Cases

- **User Stories:**

  Primary User: Malaysian SME Owner:

  - As an SME owner, I want to view my real-time ESG performance so that I can understand my current eligibility for green financing.

- As an SME owner, I want to receive alerts when supplier or external events impact my ESG score so that I can take corrective action early.
- As an SME owner, I want to simulate operational changes (e.g. supplier switching or efficiency improvements) so that I can evaluate their impact on ESG scores and financial outcomes before making decisions.
- As an SME owner, I want to understand the reasons behind ESG assessments and recommendations so that I can trust and act on the insights provided.
- As an SME owner applying for financing, I want to generate a banker-ready ESG report in PDF format so that I can support my loan application process.

Secondary User: SME Loan Officer (Financial Institution):

- As a loan officer, I want to review AI-generated ESG scores with supporting evidence so that I can make faster and more accurate financing decisions.
- As a loan officer, I want to detect greenwashing risks in SME ESG claims so that I can reduce the risk of approving misleading or inaccurate applications.
- As a loan officer, I want to access structured ESG reports and summaries so that I can streamline due diligence and maintain audit compliance.

Tertiary User: ESG Program Managers / Fintech Partners (e.g. Ant International):

- As a fintech ecosystem manager, I want to monitor ESG performance across my SME portfolio in real time so that I can evaluate sustainability progress at scale.
- As a fintech partner, I want to track measurable ESG impact (e.g. carbon reduction, job creation, financing uptake) so that I can report ecosystem-level outcomes to stakeholders.
- As a fintech partner, I want to convert SME ESG performance into credit-relevant insights so that financial institutions can make more informed lending decisions.

- **Use Cases (Main Interactions):**

  UC1: View ESG Performance

  - SME owner logs into the system and views real-time ESG score dashboard.

- The system aggregates structured and unstructured ESG data and displays an explainable score breakdown.
- User gains visibility into ESG status and financing eligibility.

UC2: Monitor ESG Risk Alerts

- The system continuously monitors supply chain and external ESG data.
- When a risk event (e.g., supplier issue or negative news) is detected, an alert is triggered.
- SME receives notification and can take corrective action early.

UC3: Simulate What-If Scenarios

- SME inputs operational changes (e.g., supplier switch or efficiency improvement).
- The system simulates ESG and financial impact using predictive models.
- Results are displayed to support decision-making before implementation.

UC4: Understand ESG Reasoning

- SME requests explanation for ESG scores or recommendations.
- The system provides factor-level breakdown and multi-agent reasoning output.
- User understands how decisions are generated.

UC5: Evaluate Green Financing Options

- SME submits a request for financing recommendation.
- The system evaluates ESG performance and risk profile.
- Outputs include loan eligibility insights and financing options.

UC6: Generate ESG Report

- SME generates ESG report for loan application.

- The system compiles ESG insights, simulations, and forecasts.
- A structured, banker-ready PDF report is produced.

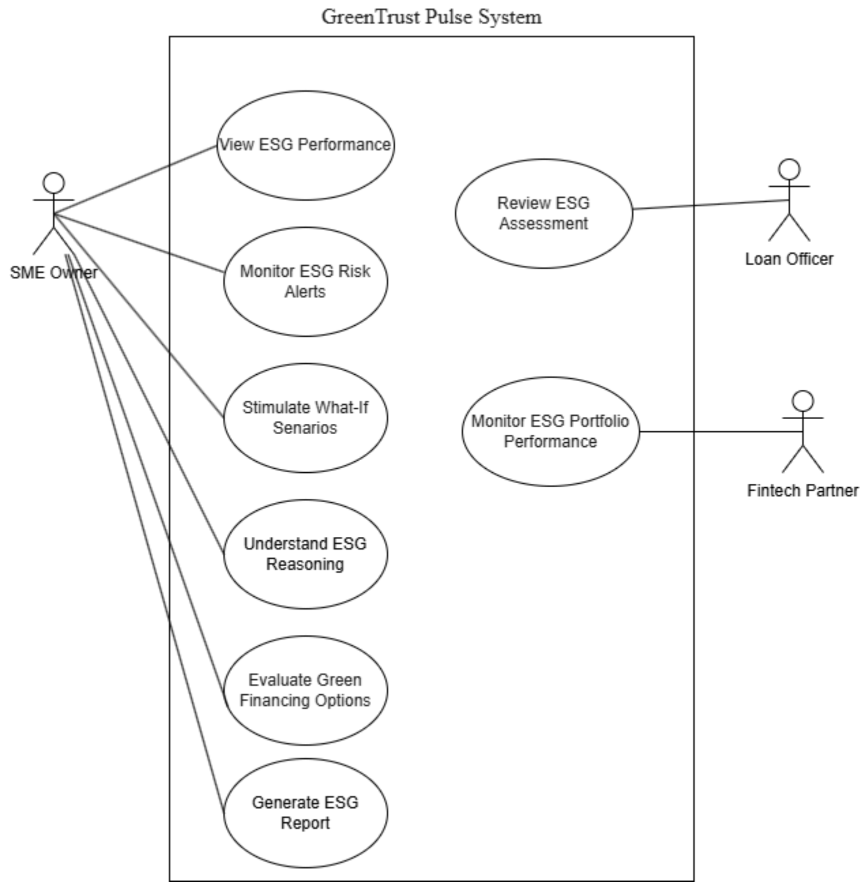
UC7: Review ESG Assessment

- Loan officer reviews ESG report and supporting evidence.
- The system highlights greenwashing risks and key ESG indicators.
- Supports faster and more accurate financing decisions.

UC8: Monitor ESG Portfolio Performance

- Fintech partner monitors ESG performance across multiple SMEs.
- The system aggregates portfolio-level ESG metrics and trends.
- Enables tracking of sustainability impact such as carbon reduction and financing uptake.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

A. ESG Monitoring & Risk Detection

- Real-time ESG monitoring and early warning system that continuously detects supply chain risks and triggers alerts via dashboard and email notifications.
- Greenwashing detection module that verifies ESG claims against external evidence such as news reports, regulatory data, and supplier records.

B. ESG Intelligence & Explainability

- ESG score computation and breakdown visualization, including factor-level explanations for environmental, social, and governance performance changes.
- Real-time visualization of multi-agent reasoning, showing ESG auditor, CFO, fraud detector, and arbitrageur debating trade-offs before a final decision, ensuring transparent and explainable outputs.

C. Decision Intelligence & Financing Support

- ESG decision intelligence engine that evaluates trade-offs between sustainability performance and financial impact to support loan eligibility and financing decisions.
- Financing recommendation system that suggests optimal funding options (e.g. green loans or sustainability-linked financing) based on ESG performance and risk profile.

D. Forecasting & Scenario Analysis

- ESG impact forecasting module that predicts future ESG scores and financial outcomes over defined time horizons (e.g. 30/60/90 days).
- What-if simulation feature allowing users to test operational or financial changes (e.g. supplier switching, efficiency improvements) and instantly observe ESG and financial impact.

E. Reporting & Export

- Banker-ready ESG report generation with one-click PDF export, consolidating all insights, decisions, and forecasts into a structured financial document.

7. Features Not Included (Scope Control)

- The system does not execute real financial transactions such as loan disbursement or payment processing, as all financing outputs are advisory recommendations only.

- The system does not perform human auditing or replace official ESG compliance verification, and all ESG scores and outputs require validation by financial institutions or relevant stakeholders.
- The system does not provide fully autonomous decision-making; final business, financing, and operational decisions remain with the user or financial institution.
- The system does not support real-time regulatory enforcement or legal certification of ESG compliance, as it is not a regulatory authority system.
- The system does not support offline operation, as continuous data ingestion and cloud-based processing are required for full functionality.
- The system does not include manual data entry as a primary workflow, since structured and unstructured ESG data are automatically retrieved through integrated data pipelines and external database connections, with user input limited to queries and scenario parameters.

8. Assumptions & Constraints

- **LLM Cost Constraint:** The system uses Z.ai GLM for multi-agent reasoning, GraphRAG-based retrieval, and ESG decision generation. An average user session involving ESG analysis, risk detection, forecasting, and financing recommendation is estimated to consume approximately 12,000–16,000 tokens per session, depending on the complexity of the query and the number of agents activated.

  To ensure inference costs remain manageable at scale, the system implements several optimisation strategies. Frequently accessed GraphRAG retrieval results (e.g. supplier profiles, ESG scores, and regulatory rules) are cached to reduce repeated token usage. In addition, lightweight inference paths are used for simple queries, while full multi-agent reasoning is only triggered for high-impact tasks such as loan evaluation, greenwashing detection, or ESG forecasting. This hybrid design significantly reduces redundant LLM calls while maintaining decision quality for critical workflows.

- **Technical Constraints:**

- The system is implemented as a prototype and does not support direct integration with live banking core systems or real financial transaction processing.
- ESG outputs (scores, forecasts, loan decisions) are generated in structured formats (JSON and PDF) and are not directly compatible with external regulatory or enterprise ESG reporting platforms without further adaptation.
- The GraphRAG knowledge graph is implemented using lightweight graph structures (NetworkX) and does not scale to distributed, enterprise-grade graph databases in its current form.
- The system supports web-based deployment (Next.js + FastAPI) and does not include native mobile applications (iOS/Android).
- Multi-agent reasoning workflows depend on Z.ai GLM inference latency, which may introduce delays under complex multi-step decision processes.

- **Performance Constraints:** The system's performance is constrained by the computational cost of multi-agent reasoning and GraphRAG-based analysis. Complex workflows such as ESG forecasting, greenwashing detection, and full agent debate (Swarm Reasoning) require multiple sequential LLM calls, which may increase response latency compared to single-step queries.

  To maintain responsiveness, real-time modules (e.g. early warning alerts) are prioritised over heavy analytical tasks, while forecasting and reasoning trace generation may have slightly delayed outputs under high system load. In addition, what-if simulations use simplified predictive models rather than high-fidelity financial simulations to ensure fast interactive feedback.

  Token-intensive outputs such as detailed reasoning traces and banker-ready ESG reports are also subject to generation limits to balance output quality with system performance and cost efficiency.

- **User Input:** The system does not require manual data entry for ESG analysis, as structured data (e.g. financial records, ESG scores) and unstructured data (e.g. news articles, ESG reports, regulatory updates) are automatically retrieved and updated through integrated data pipelines and database connections.

User interaction is primarily limited to natural language queries for analysis and decision support, as well as inputs for the what-if simulation module, where users adjust operational or financial parameters to test different scenarios.

Users may update structured business data within their own external databases, which are then reflected in the system through data synchronization. However, all generated outputs including ESG insights, forecasts, and financing recommendations remain advisory in nature and require human validation before any real-world decision-making.

9. Risks & Questions Throughout Development

- **Design Similarity:** A key risk is that the system may generate similar ESG insights, recommendations, or reasoning outputs for different users when faced with comparable inputs. This is due to the deterministic nature of structured prompts, shared GraphRAG context, and consistent multi-agent roles.

  To mitigate this, the system introduces variability through dynamic context retrieval from GraphRAG, ensuring that outputs are grounded in user-specific supply chain relationships and ESG histories. In addition, stochastic sampling parameters (e.g. temperature control in Z.ai GLM) and scenario-based inputs in the what-if module help diversify generated outputs.

- **Frame Stability:** The system ensures consistent and stable layout rendering by using predefined UI component templates and a fixed design system within the Next.js frontend. All ESG dashboards, reports, and visualization modules follow a standardized grid and component-based structure, ensuring that layouts remain consistent across different devices and screen sizes.

For exported outputs such as PDF ESG reports, the system uses structured formatting libraries to enforce fixed layout rules, ensuring that exported documents maintain consistent alignment and spacing without requiring manual user adjustment.

As a result, users do not need to fix or modify any layout manually, as both in-app views and exported reports are automatically rendered in a stable and consistent format.

- **Drawing Interpretation**: The system does not rely on free-form sketch interpretation as a core function; instead, it processes structured data inputs and natural language queries, with optional visualization-based outputs generated from predefined templates.

  If an image, diagram, or sketch is provided but cannot be reliably contextualised by the AI (e.g. unclear annotations or missing semantic structure), the system will not attempt unsafe or speculative interpretation. Instead, it will fallback to requesting clarification from the user or convert the input into a generic placeholder representation for further processing.

  This ensures that ambiguous visual inputs do not lead to incorrect ESG analysis, risk evaluation, or decision outputs, maintaining consistency and reliability in downstream multi-agent reasoning and GraphRAG-based analysis.